

Generation Interconnection System Impact Study Report

For Queue Project Q12-0094

Mcconnico 138 kV

January 19, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0094. The Interconnection Customer proposes a 151.4 MW solar photovoltaic facility interconnecting at the Mcconnico 138 kV substation in the Virginia area. The request was received on January 12, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 66 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,161,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Clifton – Colonial 138.0 kV Ckt 1	Colonial – Clifton 138.0 kV Ckt 2	108.4	336	5.2
Artesia Country Club – Ross Fork Substation 138.0 kV	Air Force Base – Artesia Country Club 345.0 kV line	103.2	231	5.4
Woods Dp – To Vaca 138.0 kV #1	Woods Dp – Toa Vaca 138.0 kV Cir. 2	101.4	319	6.5
Colonial – Clifton 138.0 kV Line 2	Clifton – Colonial 138.0 kV Line 1	101.3	360	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Clifton – Colonial 138.0 kV Ckt 1	Reconductor line	\$973,000	31% (\$305,000)
Artesia Country Club – Ross Fork Substation 138.0 kV	Reconductor line	\$1,460,000	36% (\$530,000)
Woods Dp – To Vaca 138.0 kV #1	Reconductor line	\$379,000	27% (\$104,000)
Clifton – Colonial 138.0 kV Ckt 2	Reconductor line	\$710,000	31% (\$222,000)
	Total allocated to project		\$1,161,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.